

Integrating Social Media Disinformation Dataset into Linked Data

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Abstract—This project focuses on improving the FAIRness and interoperability of a dataset about social media and misinformation. The dataset chosen, “Social Media & Misinformation Dataset 2024”, contains posts from platforms like Twitter and Facebook, along with information such as engagement metrics, sentiment, misinformation flags, and other metadata.

Originally provided as a CSV file, the dataset had a low FAIR level, especially regarding interoperability and linkage to external sources. The goal of this work was to clean, harmonise, and link the data, and then describe it with well-defined properties based on existing metadata schemas.

After defining and aligning the vocabulary, the dataset was converted into JSON-LD, making it a 5-star Linked Open Data resource. Finally, a SHACL file was created to validate the quality and consistency of the data model.

Index Terms—Linked Data, Open Data, FAIR

I. INTRODUCTION

In recent years, the proliferation of misinformation on social media has posed significant challenges to public discourse and information reliability. Datasets capturing social media activity are crucial for studying these phenomena, but they are often provided in formats that hinder interoperability and integration with external data sources. One such dataset, the “Social Media & Misinformation Dataset 2024”[1] contains posts from platforms like Twitter and Facebook, along with engagement metrics, sentiment scores, and misinformation annotations.

Originally distributed as a CSV file, the dataset exhibited low FAIRness, particularly in terms of interoperability and linkage to external resources. To address this, this project focuses on cleaning and harmonizing the data, aligning it with standard vocabularies, and linking it to external knowledge sources. By converting the dataset into JSON-LD and defining a SHACL schema for validation, the dataset becomes a fully FAIR, 5-star Linked Open Data resource, enabling more effective reuse, integration, and analysis in the study of misinformation scenarios.

II. DATA CLEANING AND LINKING

This step involved cleaning, harmonizing, and linking the dataset to external data sources.

Since the dataset contained only about 40 rows, identifying and removing irrelevant or incomplete entries was straightforward. Approximately 12 rows were discarded due to null values, resulting in a more concise dataset.

After cleaning, the data was linked to external sources. A Python script using the `pywikibot` library was developed to automatically match each cell’s value to its corresponding Wikidata entity. For example, in the platform column, the value “Twitter” was replaced with the entity identifier “Q1390577.” Columns with non-linkable information were excluded from this process.

The output of this phase was a cleaned CSV file where most cells contained Wikidata entity identifiers, except for columns with data that could not be linked, such as timestamps or numeric values.

III. PROPERTY AND TERM DEFINITION

Before defining terms for our dataset, we identified several fields without direct equivalents in existing vocabularies such as Wikidata, DCTerms, or Schema.org. To address this, we created a custom namespace on `purl.org` named `smsm` (“Seminars Social Media”). This namespace defines properties not covered by standard vocabularies, such as `platform`, `factCheckSource`, and `sentimentScore`.

Below is the RDF definition of the platform property:

```
smsm:platform
  a rdf:Property ;
  rdfs:label "Platform"@en ;
  rdfs:comment "The social media
platform on which the post was posted
on"@en ;
  rdfs:range xsd:string .
```

Now we can define terms for each of our dataset’s fields.

TABLE I
EXCERPT OF THE FIELDS INSIDE THE CHOSEN DATASET, AND THEIR RESPECTIVE TERM

Label	Term	Cardinality	Range	Syntax
Content Text	schema:text	1	xsd:string	-
Platform	smsm:platform	1	xsd:string	-
Viral Score	smsm:viralScore	1	xsd:decimal	-
Post ID	schema:identifier	1	xsd:string	“ig_024”
Engage-ment	smsm:engagement	1	xsd:decimal	-

IV. CONVERSION TO SERIALIZATION LANGUAGE

The original dataset was provided as a CSV file, which, according to Tim Berners-Lee's Open Data rating, corresponds to a 3-star level. To enhance interoperability, the data was converted to JSON-LD using a Python script built with the `rdflib` library.

Below is an excerpt of a JSON-LD object representing a social media post after conversion from the original CSV file:

```
"@id": "https://www.kaggle.com/datasets/imaadmahmood/social-media-and-misinformation-dataset-2024/linked_dataset/tw_001",
"@type": "http://schema.org/SocialMediaPosting",
"schema:about": [
  {
    "@id": "wd:Q7937"
  },
  {
    "@id": "wd:Q192864"
  },
  {
    "@id": "wd:Q2249676"
  }
],
"schema:identifier": "tw_001",
"schema:datePublished": {
  "@type": "xsd:dateTime",
  "@value": "2024-08-13T09:15:23Z"
},
"smsm:platform": {
  "@id": "wd:Q1390577"
},
},
```

V. DEFINITION OF SHACL

Finally, the SHACL file was written to validate the structure and consistency of the dataset once it was converted into RDF. The goal was to ensure that each entity representing a social media post complied with the expected data model and semantic constraints. To achieve this, a `NodeShape` was defined for the class `schema:SocialMediaPosting`, targeting all instances of that type. Each relevant field from the dataset was then described using `sh:property` constraints to specify the expected datatype, cardinality, and, when appropriate, node kind (e.g., IRI or literal).

Standard Schema.org terms were used whenever possible (such as `schema:identifier`, `schema:datePublished`, and `schema:about`), while custom properties from the project's namespace (e.g., `smsm:platform`, `smsm:sentimentScore`, `smsm:factCheckSource`) were included to capture dataset-specific attributes. These constraints help detect missing, incorrectly typed, or semantically invalid data.

Additionally, a separate shape, `schema:InteractionCounterShape`, was defined for nested interaction statistics. This ensures that each interaction record includes a valid interaction type and count.

VI. CONCLUSIONS

In conclusion, this work demonstrated the process of transforming a simple CSV dataset into a structured, interoperable, and semantically rich format using Linked Data principles. Through data cleaning, harmonization, and linking to external sources such as Wikidata, the dataset achieved higher FAIRness and reusability. Custom terms were defined in the `smsm` namespace to address missing vocabulary coverage, and the resulting RDF data was validated with a SHACL schema to ensure structural consistency.

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REFERENCES

- [1] I. Mahmood, "Social Media and Disinformation 2024." [Online]. Available: <https://www.kaggle.com/datasets/imaadmahmood/social-media-and-misinformation-dataset-2024>